

JKBOSE 2026

CLASS 11TH

#XUV 

HYDROCARBONS

LECTURE-08 CHEMISTRY

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JK Bole Wallah

Topics To Covered

- 1 Structure of Benzene**
- 2 Aromaticity**
- 3 Preparation of Benzene**
- 4 Physical Properties**
- 5 Chemical Properties**
- 6 Directive Influence of a Functional**

TIGAR Aaj LAST
CLASS NAHI

HA 😊

REVISION

Structure of ethyne.

Prep. of ethyne.

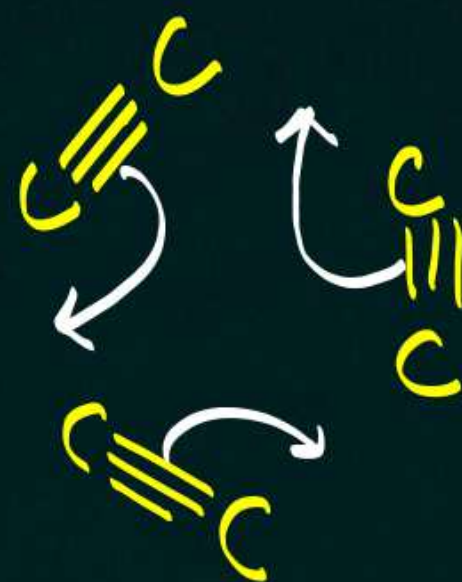
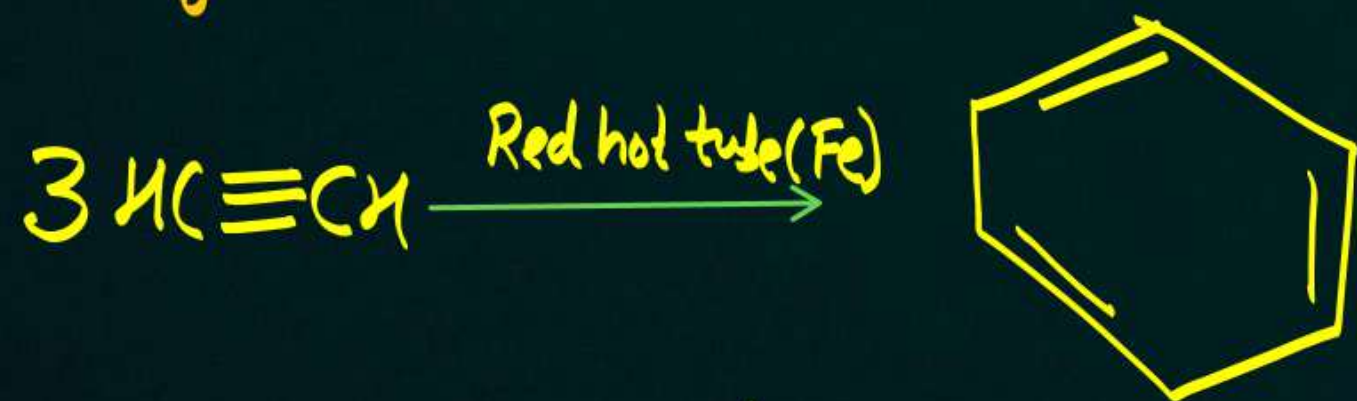
Chemical Properties.

BUNAICHARA
ON
TOP

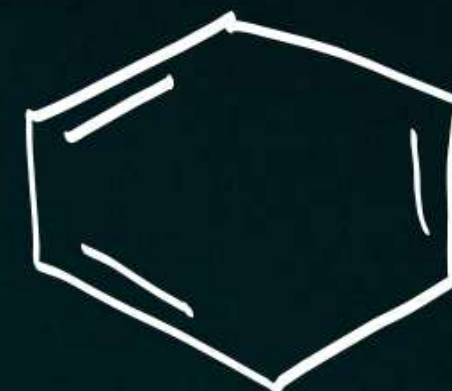
ALKYNES

c) Polymerisation :-

If we pass ethyne from the red hot tube (Fe) it forms benzene ring



Red hot tube



Benzene

ALKANES, ALKENES, ALKYNES.

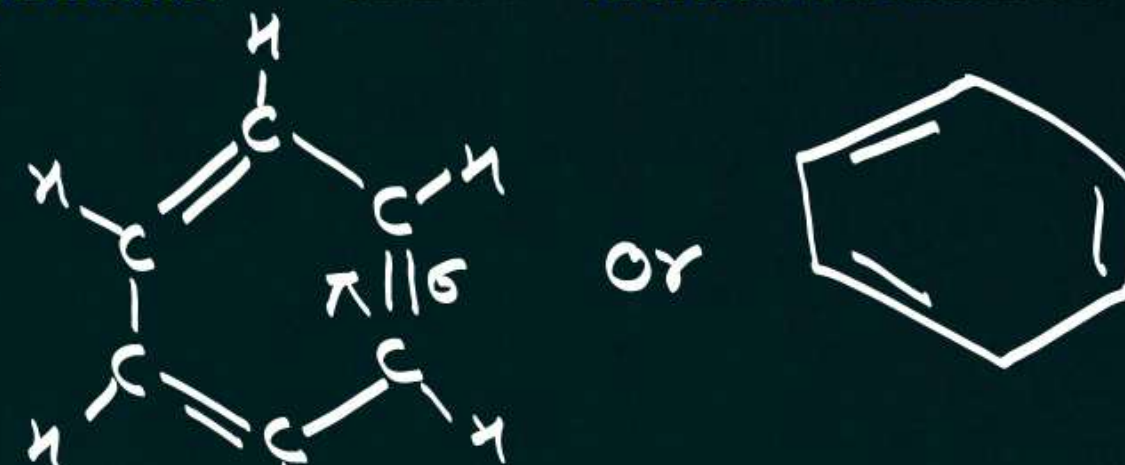
PART - IV Aromatic Hydrocarbon



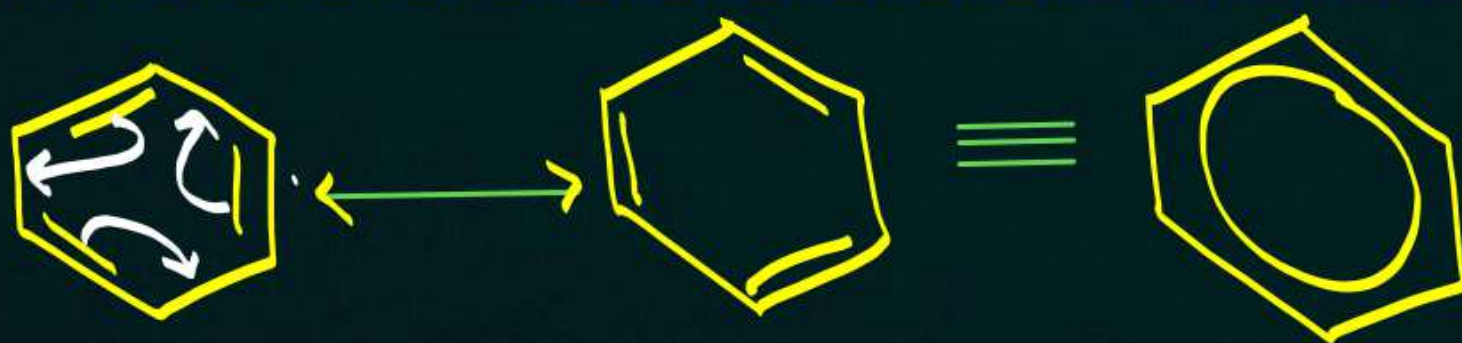
Structure of Benzene



- Benzene was first discovered by Michael Faraday in 1825. Its formula is C_6H_6 , which shows high unsaturation, yet it is very stable.



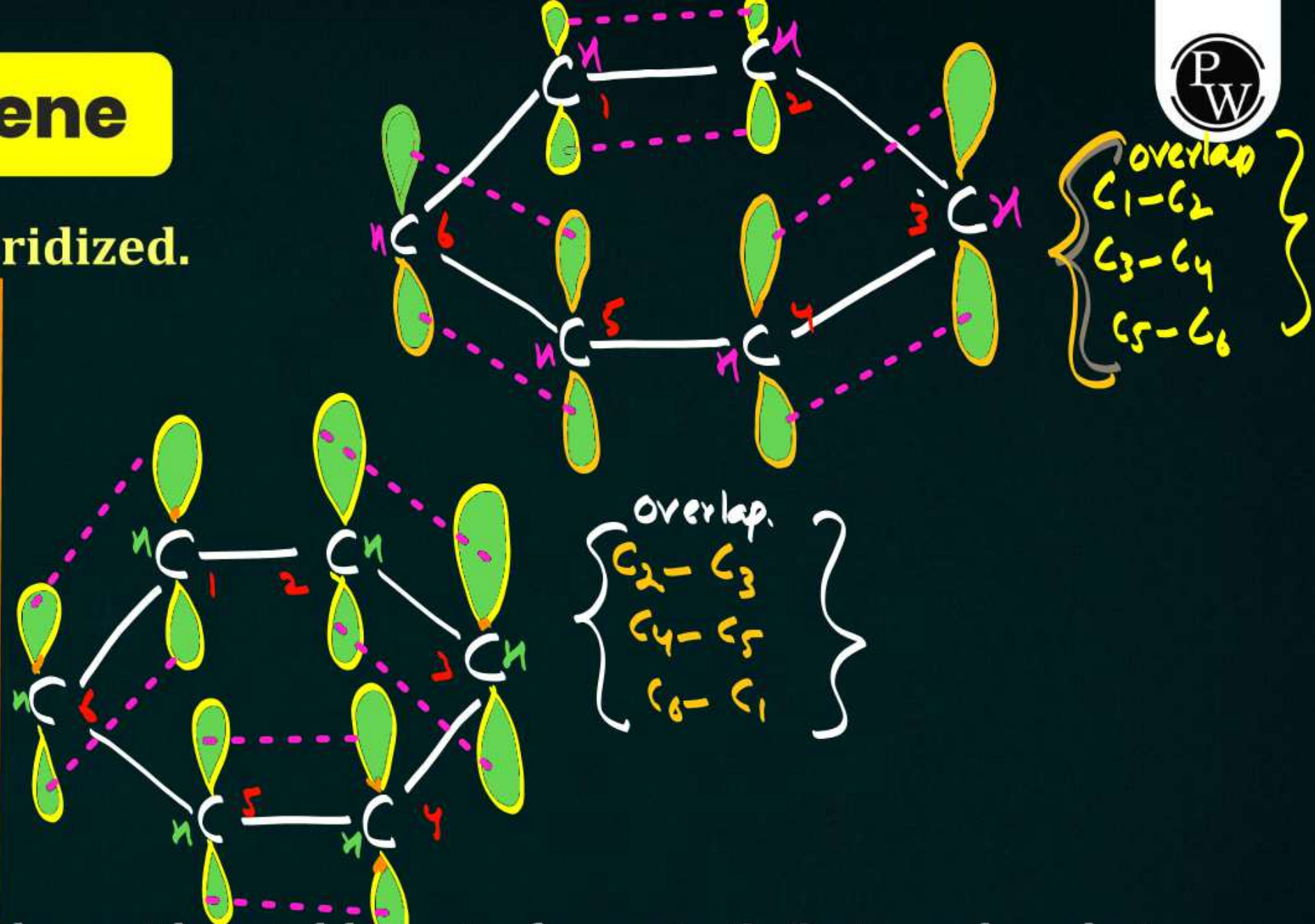
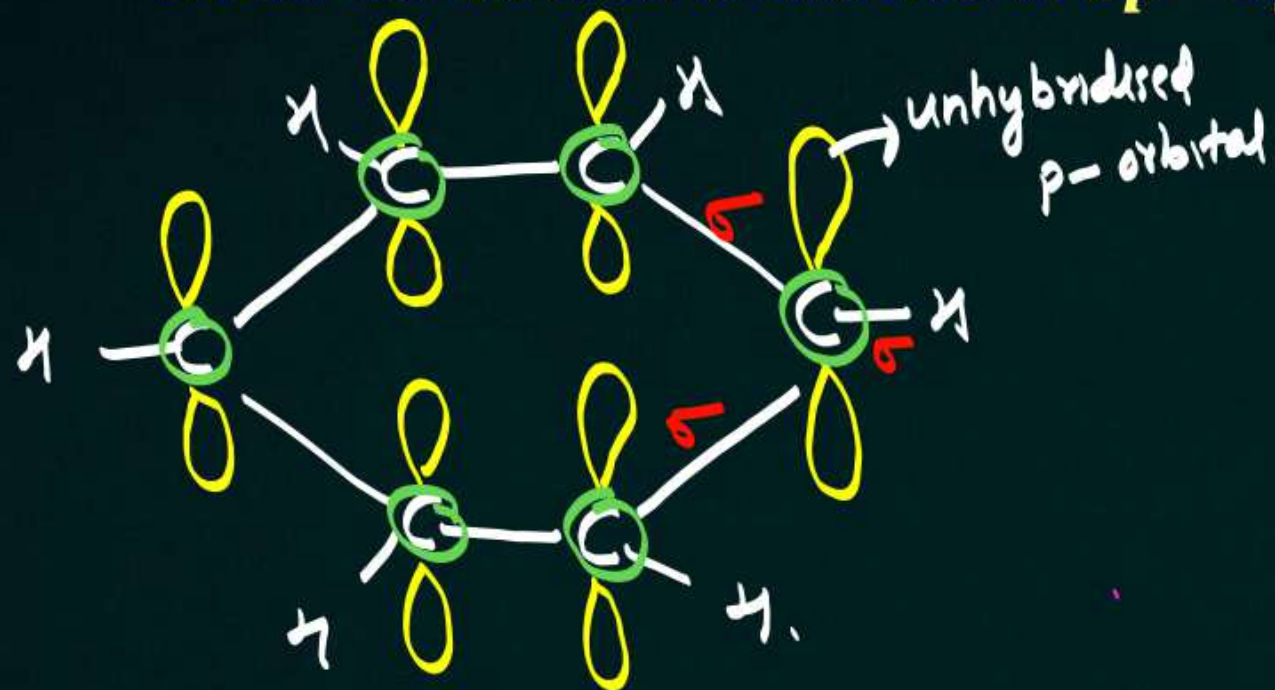
- In 1865, August Kekulé proposed that benzene has a ring of six carbon atoms with alternate single and double bonds, each carbon attached to one hydrogen.



- The stability of benzene is explained by resonance. It is not just one structure but a resonance hybrid of two forms. The six electrons are delocalized over the ring, shown by a circle inside the hexagon, which gives benzene its extra stability.

Orbital Structure of Benzene

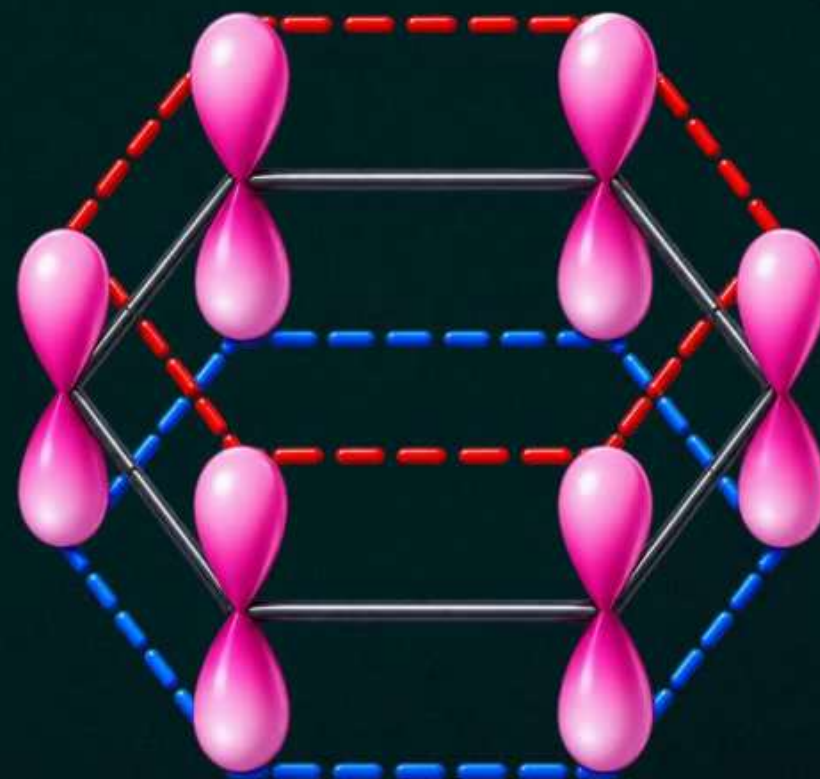
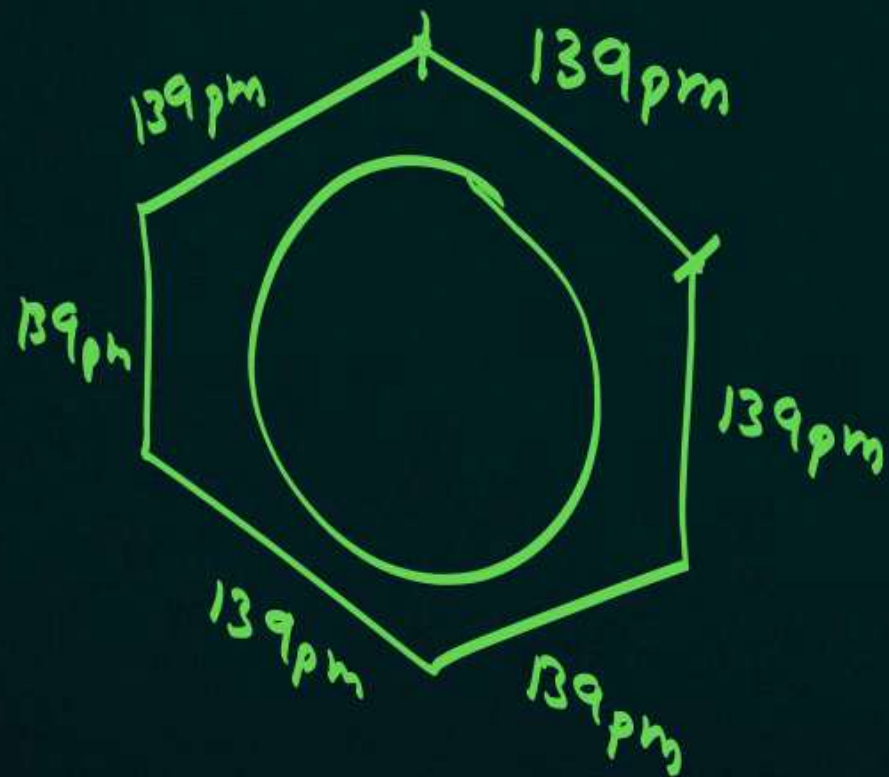
- Each carbon in benzene is sp^2 -hybridized.



- Two sp^2 orbitals of each carbon overlap with neighbors to form six C-C sigma bonds in a hexagonal ring.
- The third sp^2 orbital overlaps with hydrogen's 1s orbital to form six C-H sigma bonds.
- Each carbon also has one unhybridized p-orbital, perpendicular to the ring plane. These p-orbitals overlap sideways with adjacent ones, creating a delocalized π -electron cloud.

Orbital Structure of Benzene

- X-ray diffraction shows all C–C bonds are equal (139 pm), intermediate between single and double bonds. The delocalized electrons form two continuous clouds: one above and one below the benzene ring, giving extra stability.



or



Aromaticity (Most Important).



A compound can be considered as aromatic if it follows following three rules.

- (i) An aromatic compound must be cyclic and planar.
- (ii) An aromatic compound must have the conjugated system in it.
- (iii) An aromatic compound must follow Huckel rule which states that the number of π -electrons = $(4n + 2)$; where n = integer 0, 1, 2, 3 etc.

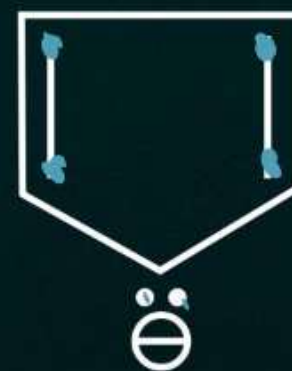
- 1) Cyclic ✓
- 2) Conjugation ✓
- 3) $(4n+2)\pi e^-$ ✓



Aromatic
(2π)



Aromatic
($6\pi e^-$)



Aromatic
($6\pi e^-$)



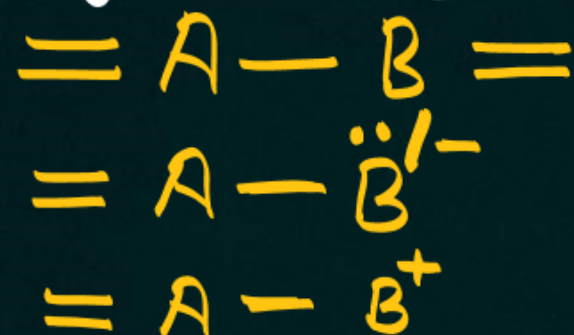
Aromatic
($6\pi e^-$)

- 1) Rabia Mushtaq
- 2) Jahangir Hussain
- 3) War Safoora Javid.

AROMATIC

1) CYCLIC

2) Conjugated System (complete).



3) $(4n+2) \pi e^-$ ($n = 0, 1, 2, 3, \dots$)

$$4 \times 0 + 2 = 2\pi e^- \quad \checkmark$$

$$4 \times 1 + 2 = 6\pi e^-$$

$$4 \times 2 + 2 = 10\pi e^-$$

$$4 \times 3 + 2 = 14\pi e^-$$

$$4 \times 4 + 2 = 18\pi e^- \dots \dots$$

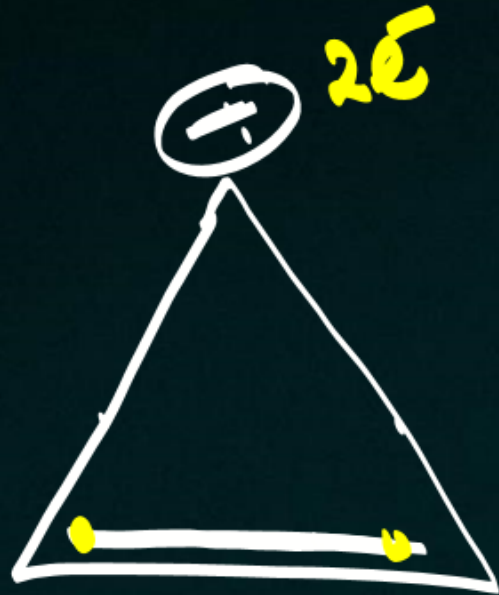
ANTIAROMATIC

Non
Aromatic



4n Rule
 πe^-

Q.



Antiaromatic

$4\pi e^-$



$oo \rightarrow 2e^-$

$- \rightarrow 2e^-$

Cyclic ✓

Conjugation ✓

$(4n+2)\pi e^-$

2, 6, 10, 14.

cyclic
conjugation
 $4n+2$



Pyrrole

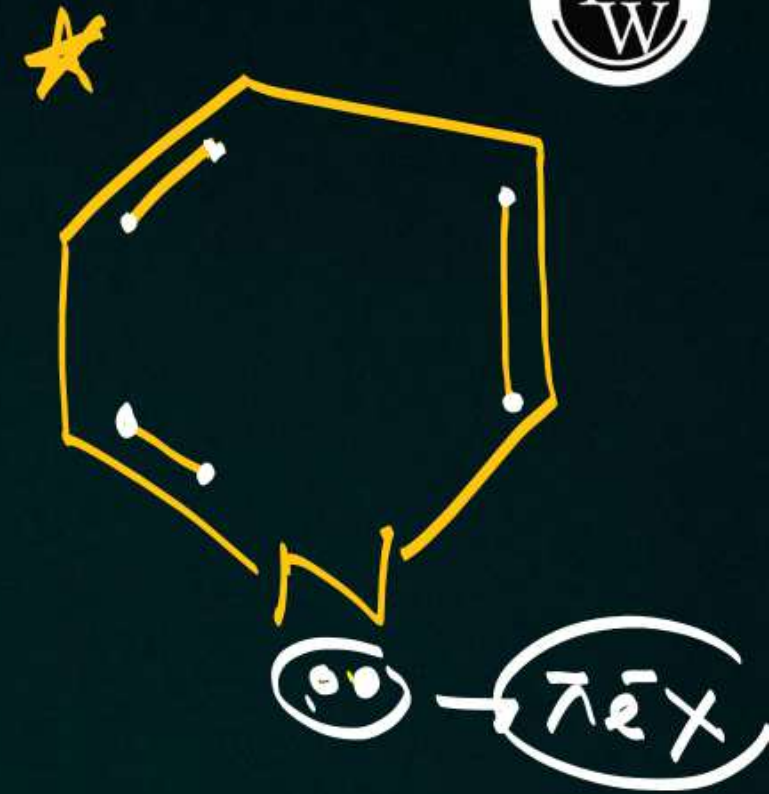
$6\pi e^-$



Furan.



Thiophene

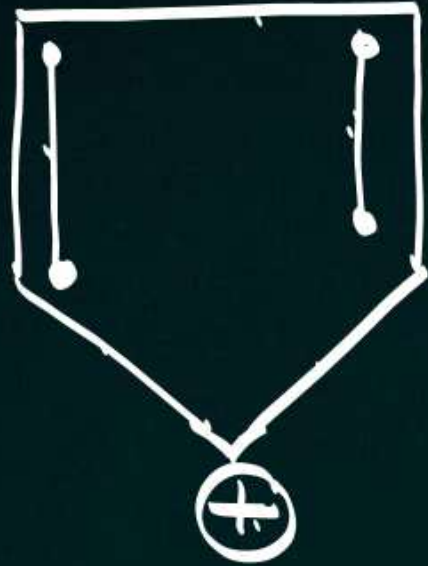


Pyridine

$6\pi e^-$

AROMATIC

Q.

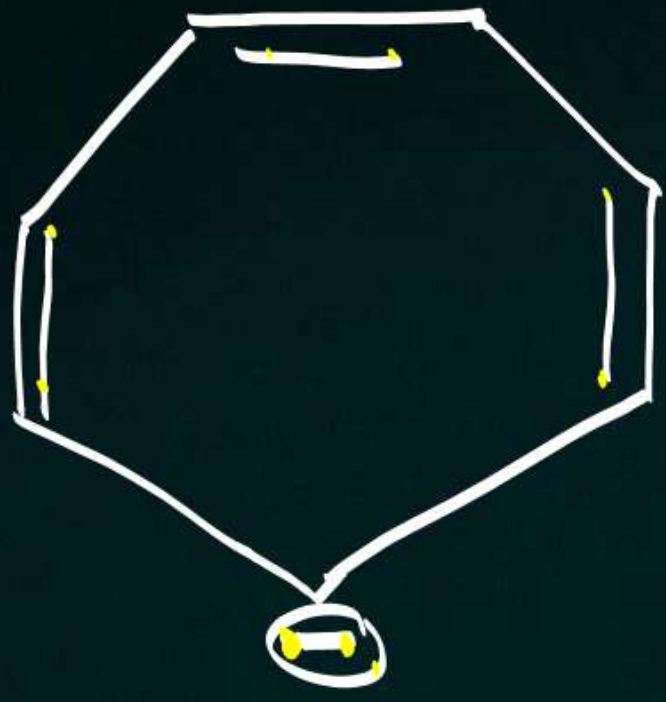


- a) Aromatic
- b) Non-Aromatic
- c) Antiaromatic
- d) None of the above.

Aromatic	Antiaromatic
→ cyclic	✓
→ Conjugation	✓
$(4n+2) \pi e^-$ → $2\pi e^-, 6\pi, 10\pi, 14\pi, 18\pi, 22\pi$	$4n \pi e^-$



Q.



-ve charge
↓
 $2e^-$

AROMATIC

Cyclic

Conjugation

$$(4n+2)\pi e^-$$

2, 6, 10, 14, 18, 22

Antiaromatic

✓

✓

$$4n$$

$n=1, 2, 3, \dots$

a) Aromatic b) Non-Aromatic

c) Antiaromatic d) None.

Q.



- a) Aromatic b) Antiaromatic
 c) Non-Aromatic d) None.

Aromatic

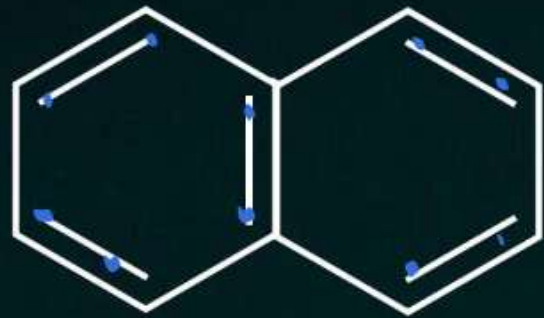
Cyclic
 Conjugation

$$4n + 2 \pi e$$

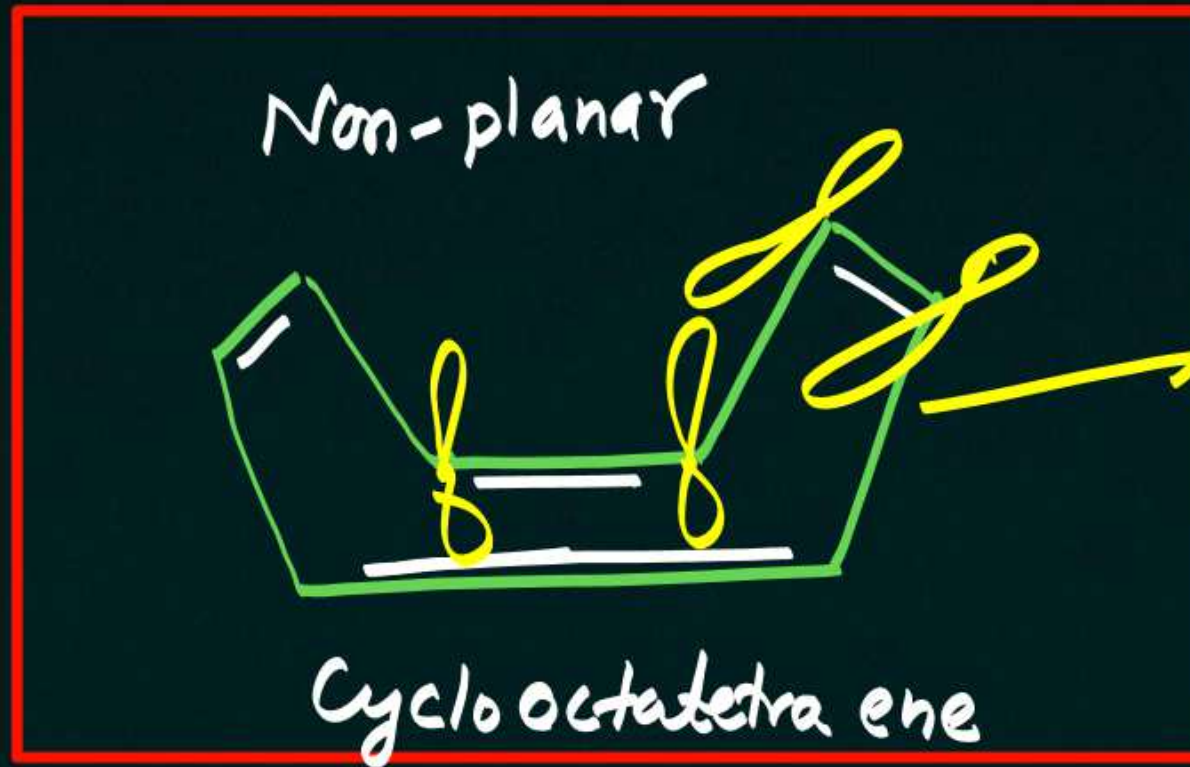
Antiaromatic

Cyclic
 Conjugation

$$4n \pi e$$



Naphthalene
(Aromatic $10\pi e$).



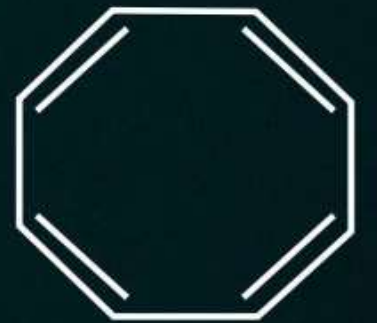
- 1) Cyclic
- X 2) Conjugation
- 3) $(4n+2)\pi e$



(Aromatic)
Anthracene



Aromatic
Phenanthrene.



Antiaromatic X
Non-Aromatic

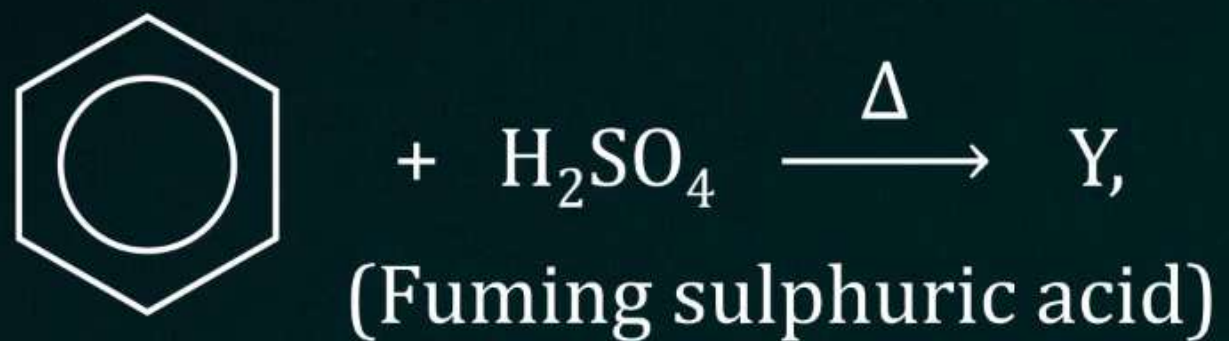
QUESTION

Which of the following set of functional groups is *meta-directing* group?

- 1 $-\text{NO}_2$, $-\text{NH}_2$, $-\text{COOH}$, $-\text{COOR}$
- 2 $-\text{NO}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COR}$
- 3 $-\text{CN}$, $-\text{CHO}$, $-\text{NHCOCH}_3$, $-\text{COOR}$
- 4 $-\text{CN}$, $-\text{NH}_2$, NHR , $-\text{OCH}_3$

QUESTION

In the reaction,



QUESTION

The aromatic compound among the following is



Thank
You